

Electricity within the body



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Academic year: 2025-2026*

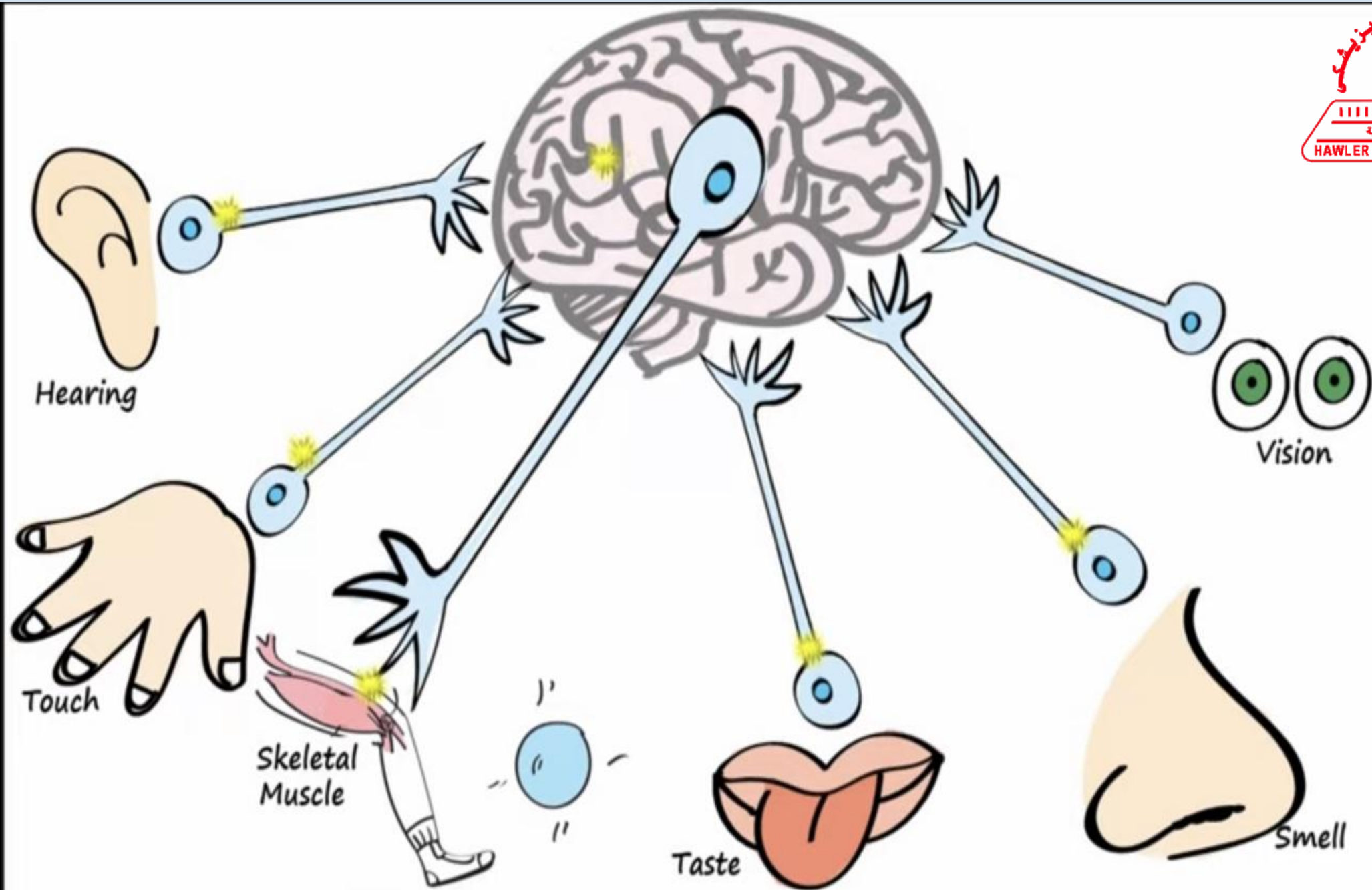
Objectives:

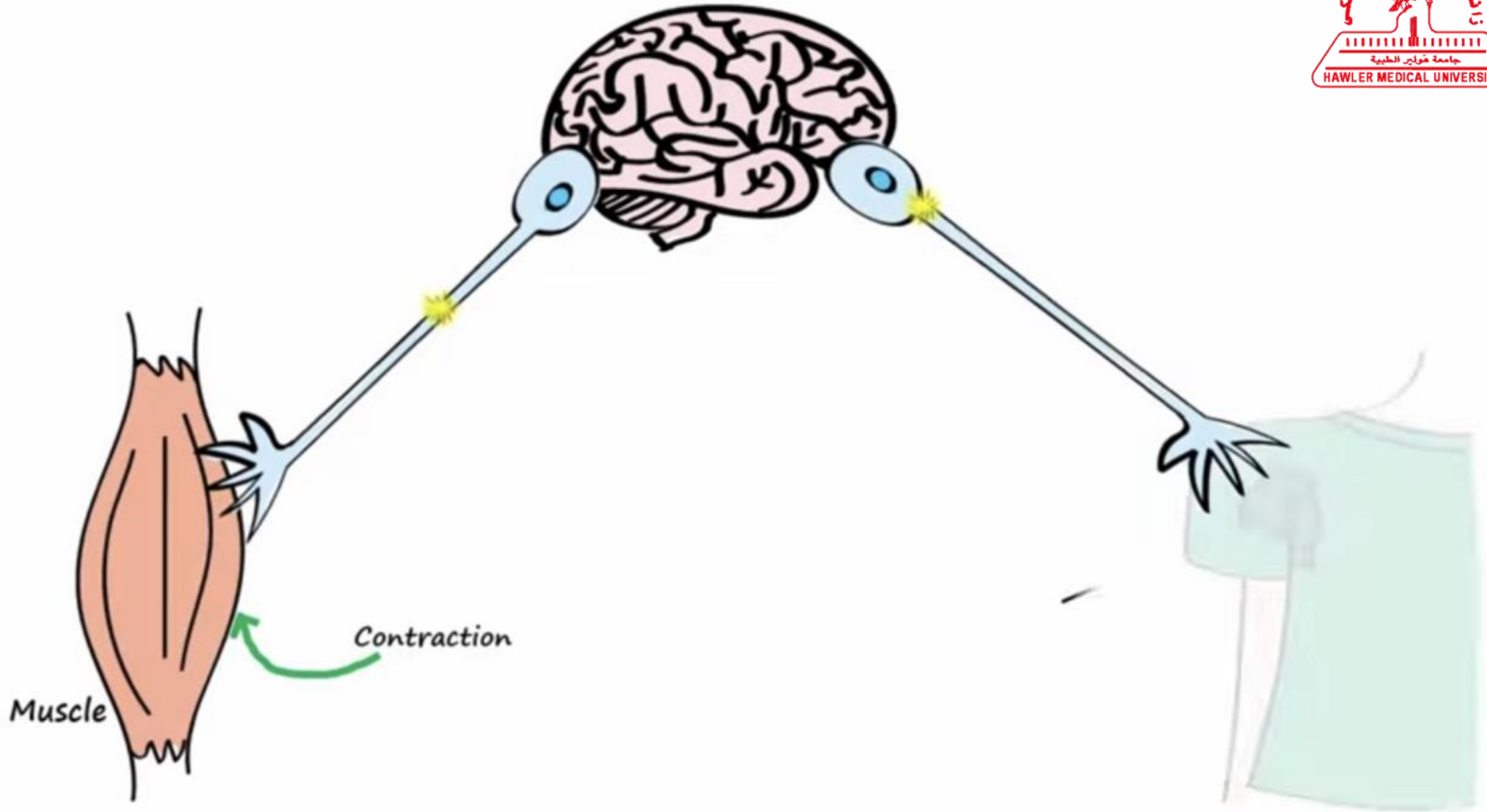
ELECTRICITY AND MAGNETISM play an important role in medicine.

- ❑ Studying generation of electrical signals inside the body.
(control and operation of nerves, muscles, and organs).

- ❑ By selectively measuring the desired signals without disturbing the body, obtaining useful clinical information about particular body functions
{NCV, EMG, ECG, EEG, EOG,ERG}

- ❑ Applying the desired electrical and magnetic pulse to the body when natural pulse failed. (pacemaker, defibrillator).





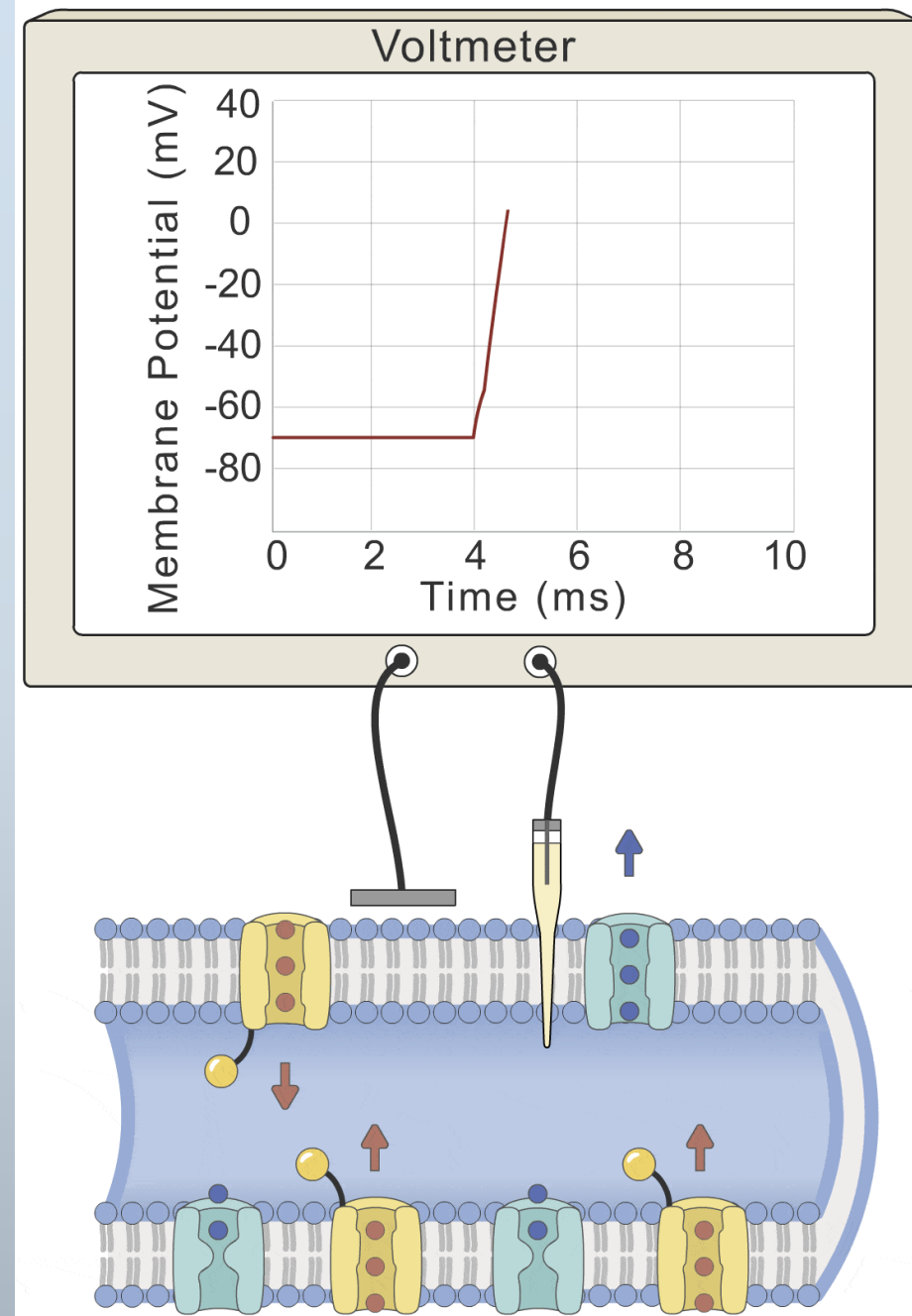
What are Biopotentials

Biopotential:

An electric potential that is measured between points in living cells, tissues, and organisms, and which accompanies all biochemical processes.

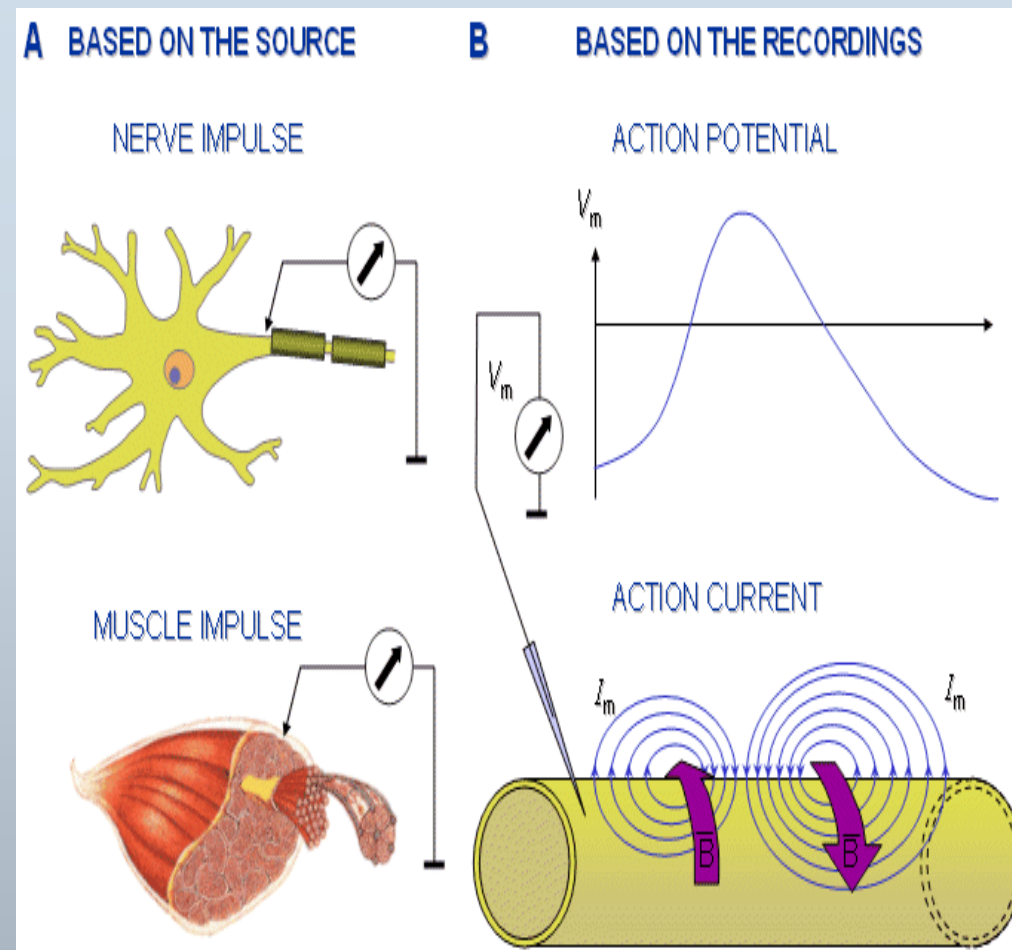
& also describes the transfer of information between and within cells

- Ionic voltages produced as a result of the electrochemical activity of *excitable cells*.
- Using transducers to convert ionic potentials into electrical potentials.



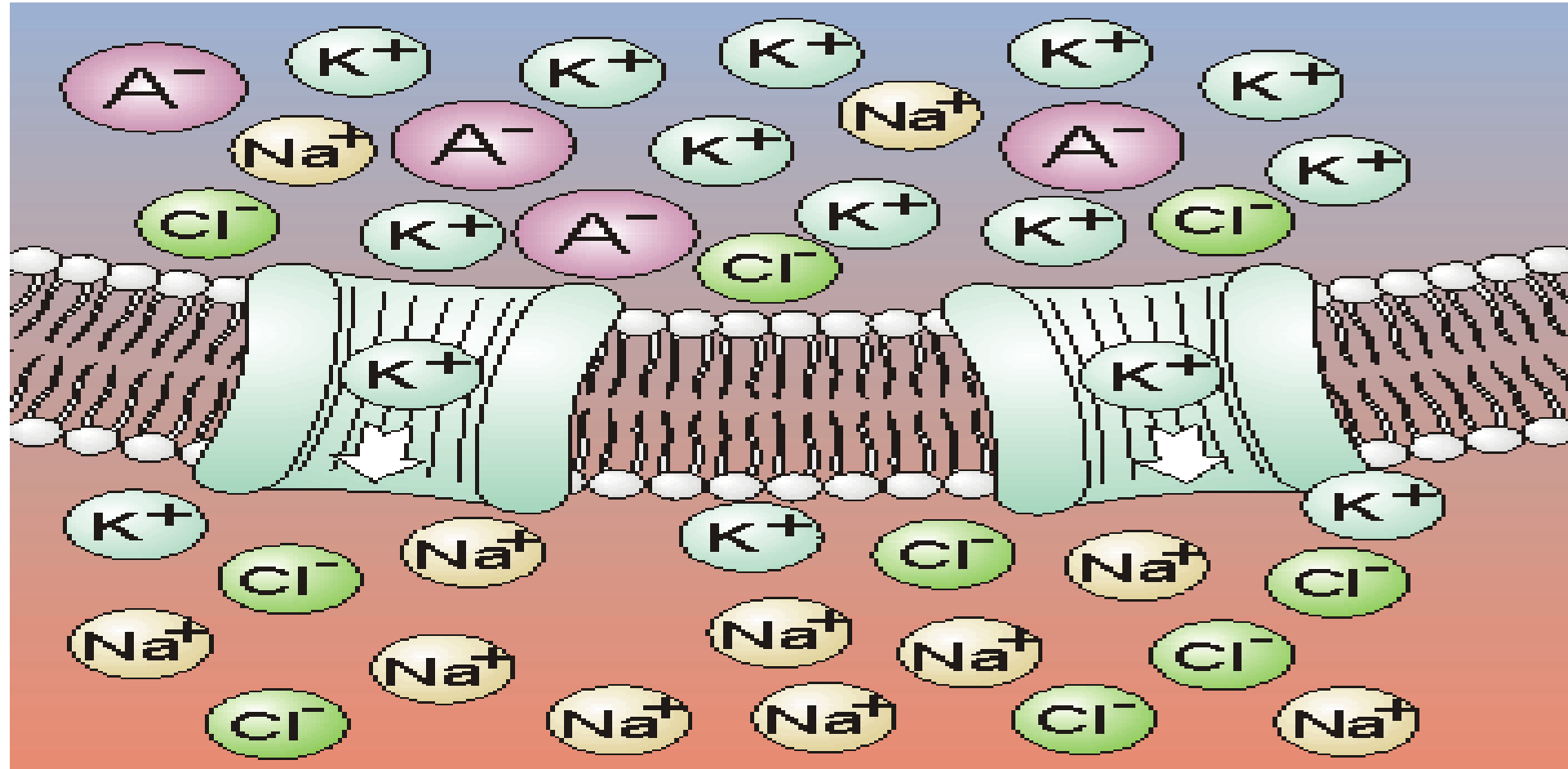
Excitable Cells

- Are components of nervous, muscular or glandular tissue
- Can produce bioelectric potentials as a result of electrochemical activity.



INTRACELLULAR MEDIUM

Cl^- Na^+ A^- K^+

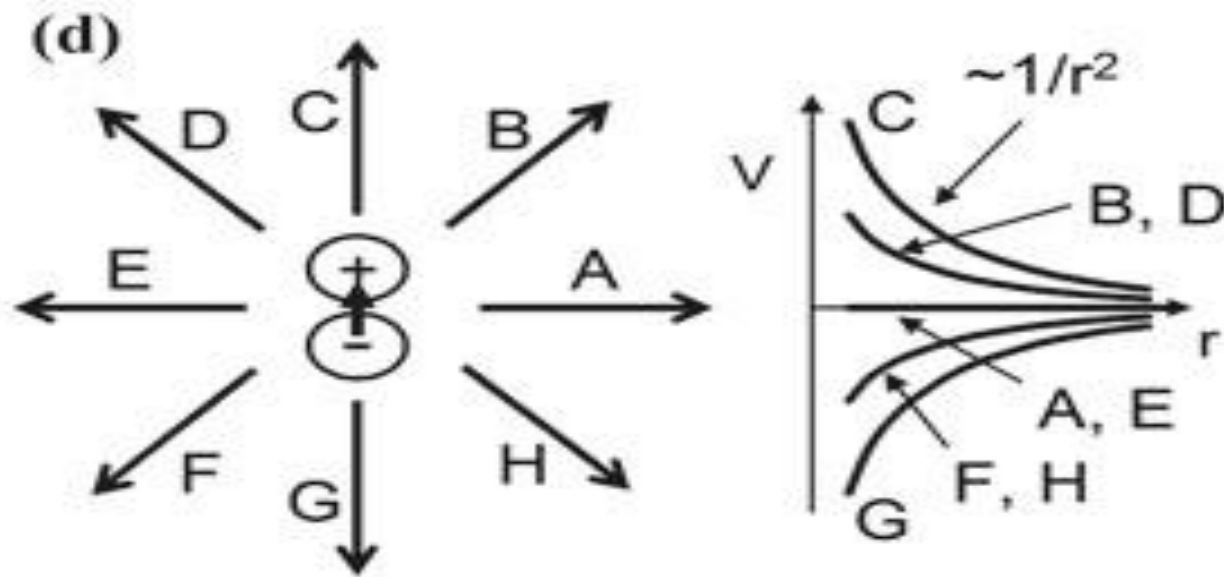
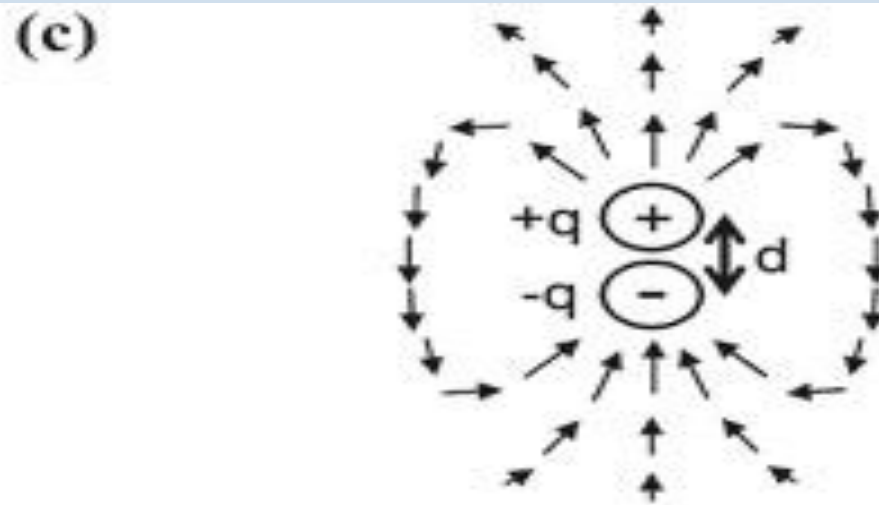
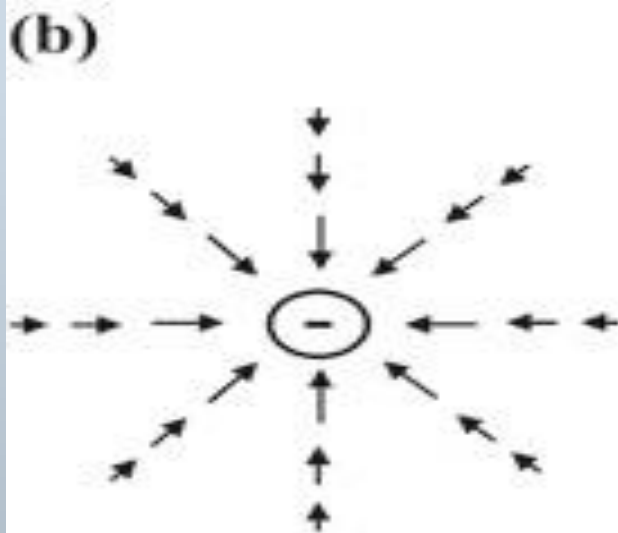
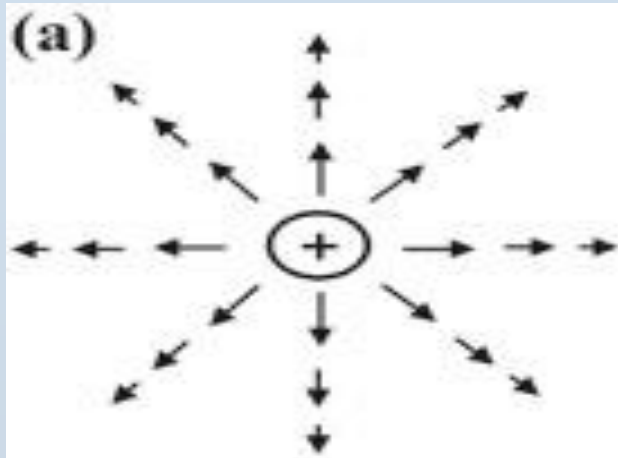


K^+ Cl^- Na^+

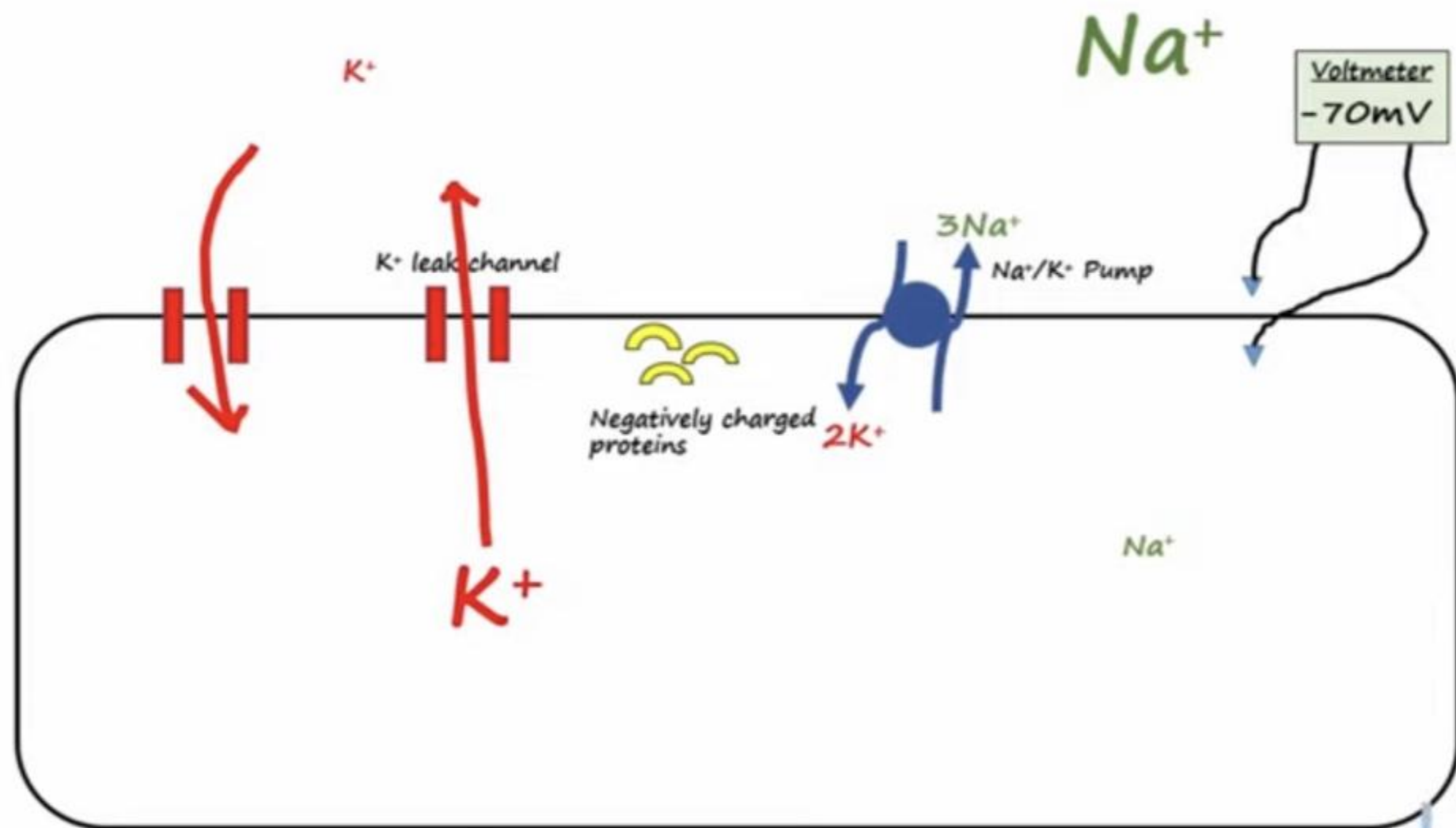
EXTRACELLULAR MEDIUM

<u>Inside axon</u>	<u>Membrane</u> ↓		<u>Extracellular fluid</u> n_i/n_o	
$[Na^+] = 15$	-	+	$[Na^+] = 145$	0.10
$[K^+] = 150$	-	+	$[K^+] = 5$	30
	-	+	$[Misc^+] = 5$	
$[Cl^-] = 9$	-	+	$[Cl^-] = 125$	0.07
$[Misc^-] = 156$	-	+	$[Misc^-] = 30$	5.2
$V = -70 \text{ mV}$	-	+	$V = 0 \text{ mV}$	
Charge neutrality	-	+	Charge neutrality	





1. **Ion concentration**
2. **Ion channels** (Na, K, Ca channel) serves as transmitters of the electric signal through the nerve
3. **Ion membrane permeability**
4. **Excitability of the cell membrane** (electrically excited by rapid changes in membrane potential due to change in ion fluxes through the membrane)
5. **Equilibrium potential** it is the membrane potential for a given ion at which electrical potential and concentration are balanced
($K^{+} = -90 \text{ mV}$, $Na^{+} = + 65 \text{ mV}$)



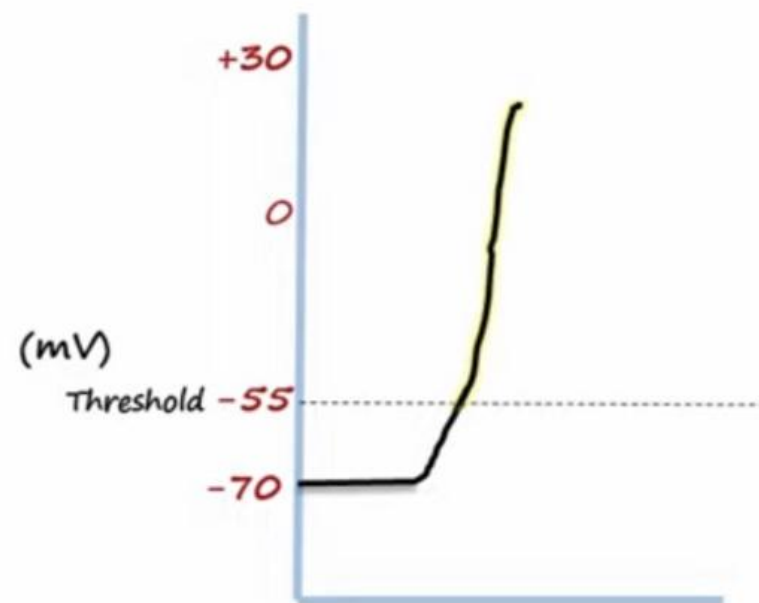
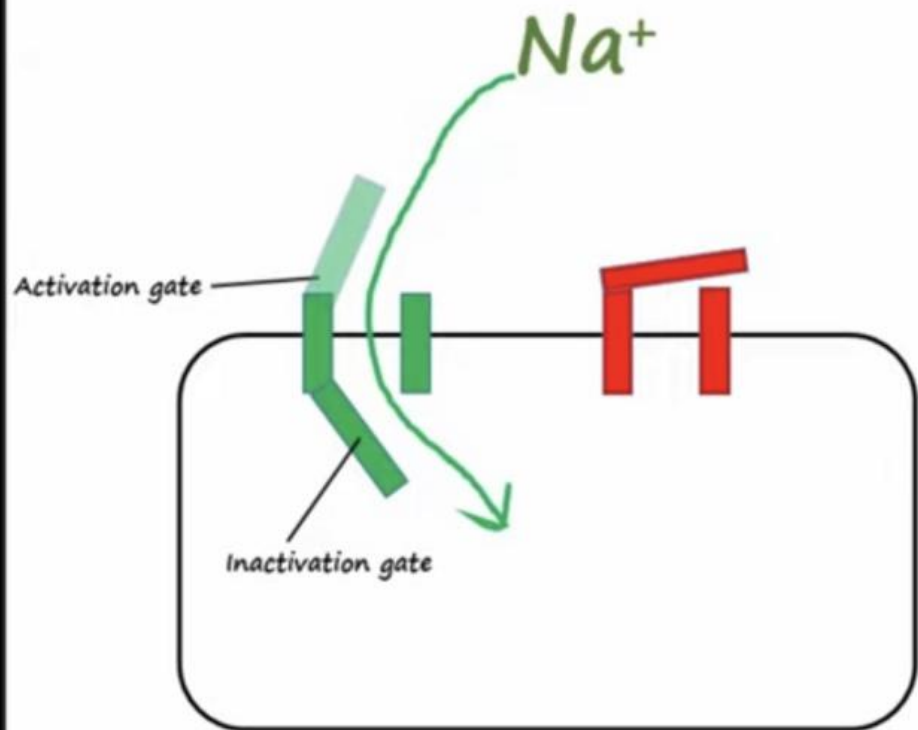
electro chemical gradient

Equilibrium is reached

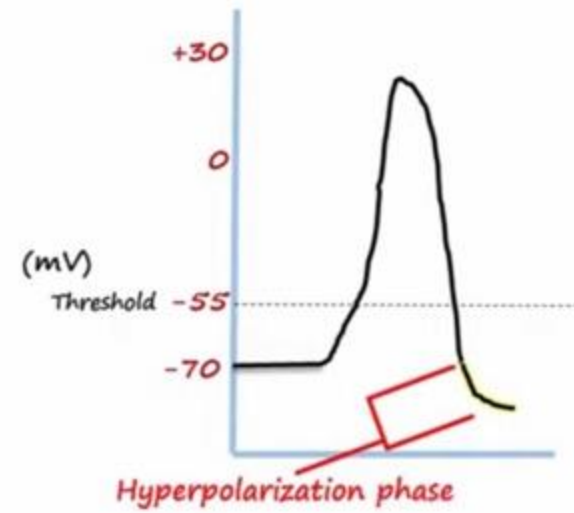
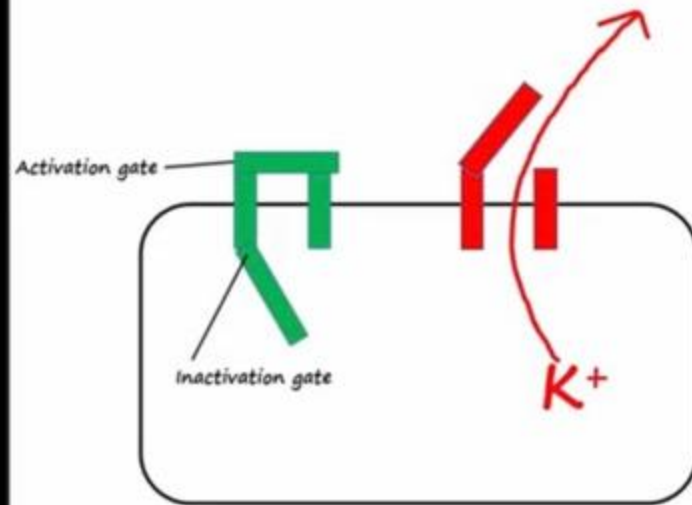
Resting Membrane Potential (RMP)

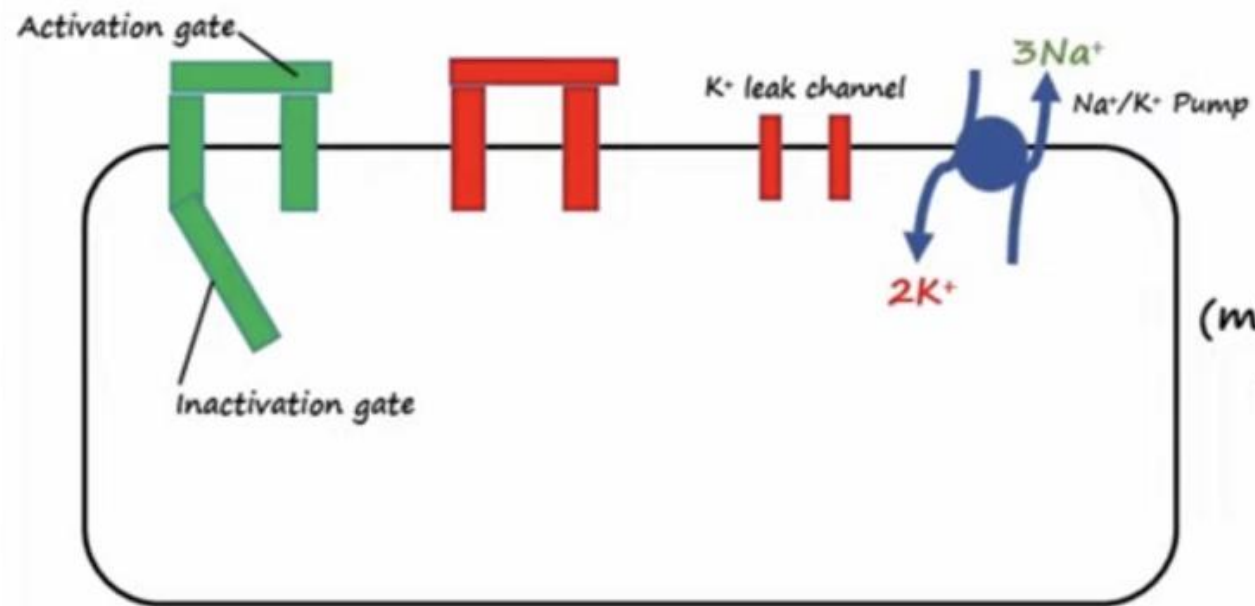
mV
-70

Closed



Act





(mV)



Graded potential

Is a short-lived local change in the membrane potential, confined to a relatively small region of the plasma membrane, and dies out within 1-2 mm of its site of origin.

Decrease with distance

Magnitude directly varies with stimulus strength

----Depolarize (-70 mV to -65)

, or

---Hyperpolarize (-70 mV to -75mV)

Action potential

Non graded rapid rise and fall of the electric potential of a cell membrane associated with the passage of impulse along the cell membrane.

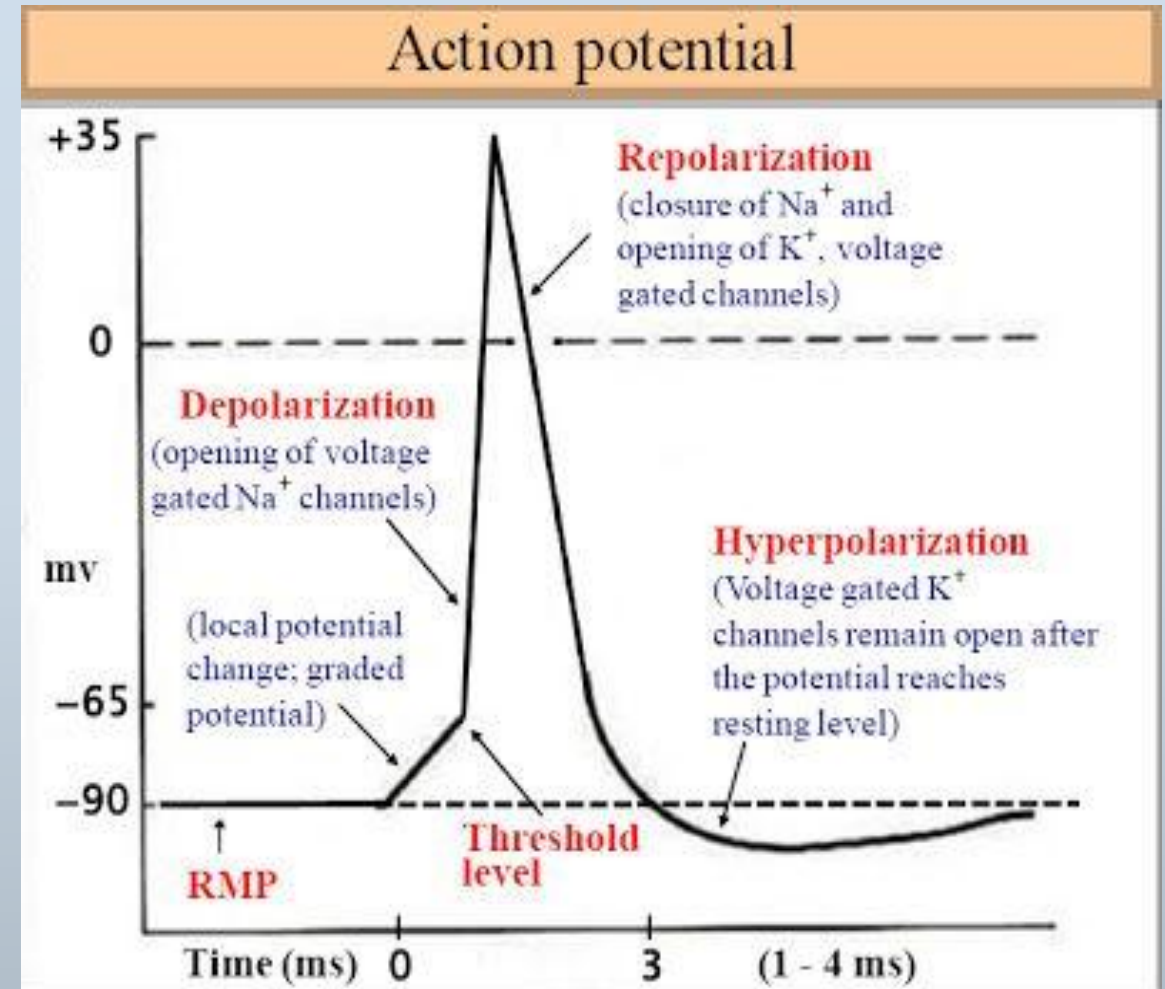
- Does not decrease with distance
- Not depend on stimulation strength
- Depolarization (-70 To +30 mV)

Follow

- Repolarization +30 mV to -70 mV
- Whole event is over in few mseconds.

STAGES OF ACTION POTENTIAL

- Resting State
 - Both sodium and potassium channels are closed (voltage-gated channels)
- Threshold
 - Stimulus opens some Na^+ channels
 - If threshold is achieved (-50mV), more Na^+ are opened triggering an action potential
- Depolarization
 - Na^+ channels are open but K^+ channels remain closed
 - Na^+ ions rush into the interior of the cell making it more positive
- Repolarization
 - Na^+ channels closed, K^+ open
 - K^+ ions leave the cell
 - Inside of cell become more negative
- Undershoot
 - K^+ channels remain open
 - The cell becomes hyperpolarized (membrane potential falls below normal resting potential)



Characteristics of action potential

1. Propagation with the same size and shape along the whole length of a nerve axon or muscle cell.
2. When several action potentials are generated due to a certain stimulation, all of them are generated in the same way.
3. Strong stimuli generate more impulses/time than that of the weak stimuli
4. When an action potential is generated, there is no other action potential can be generated (refractory period)
example : Heart = 250 msduring this period no other heartbeat can occur

Propagation of action potentials

- When an action potential is generated, it does not travel along the membrane, but it serves as a stimulus that depolarizes the adjacent membrane.
- It produces local currents of its own , depolarizes the region adjacent to it, produces another action potential at the next site
- The propagation of an action potential is the mechanism used by the nerves system to communicate over long distances.

Nerves system

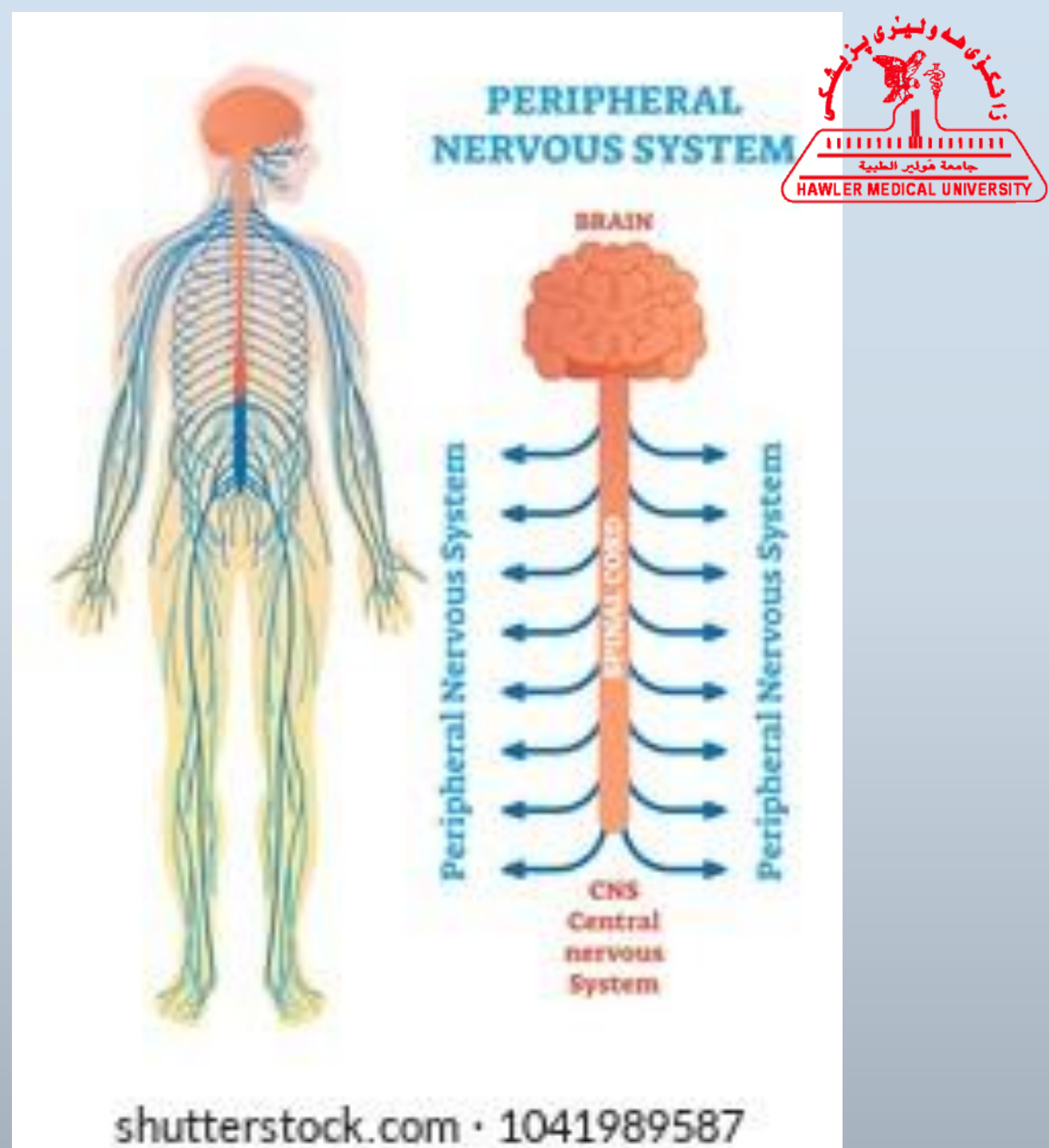
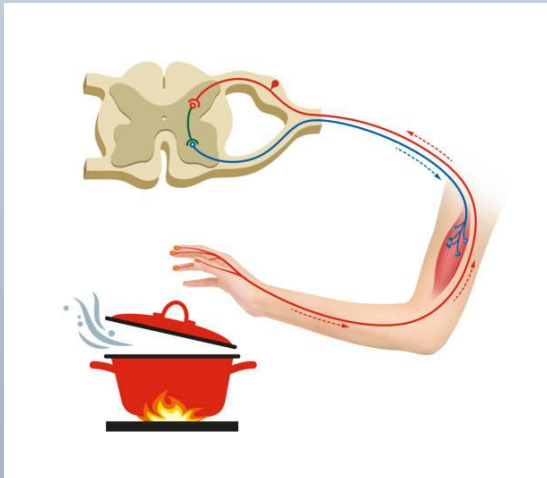
is divided into the **central nervous system** (CNS)

which is the **brain** and **spinal cord**

and the

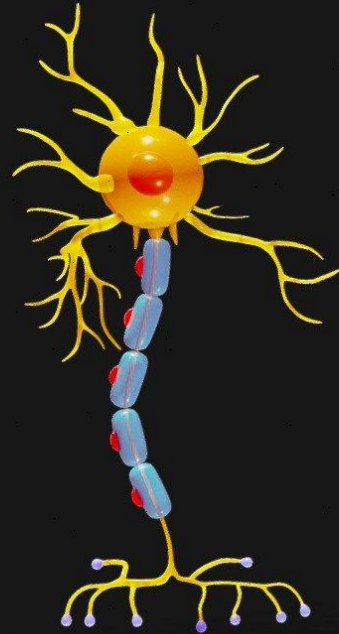
peripheral nervous system (PNS)

which **connects** everything to the brain and spinal cord

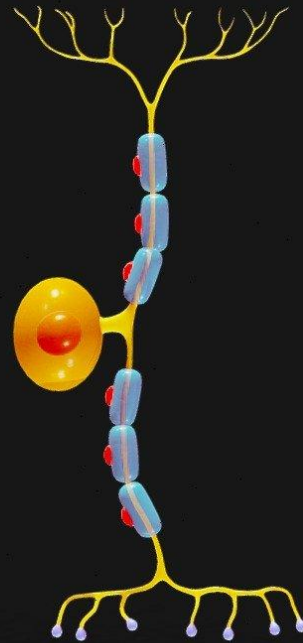


- ❑ Sensory nerve (afferent) (neurons whose axons travel from sensing area to the spinal cord)
- ❑ Motor neuron (efferent) (convey signals away from the central NS)
- ❑ 20 sensory neurons for every motor neuron
- ❑ And several thousand central processing neurons for every input or output processing

TYPES OF NEURONS



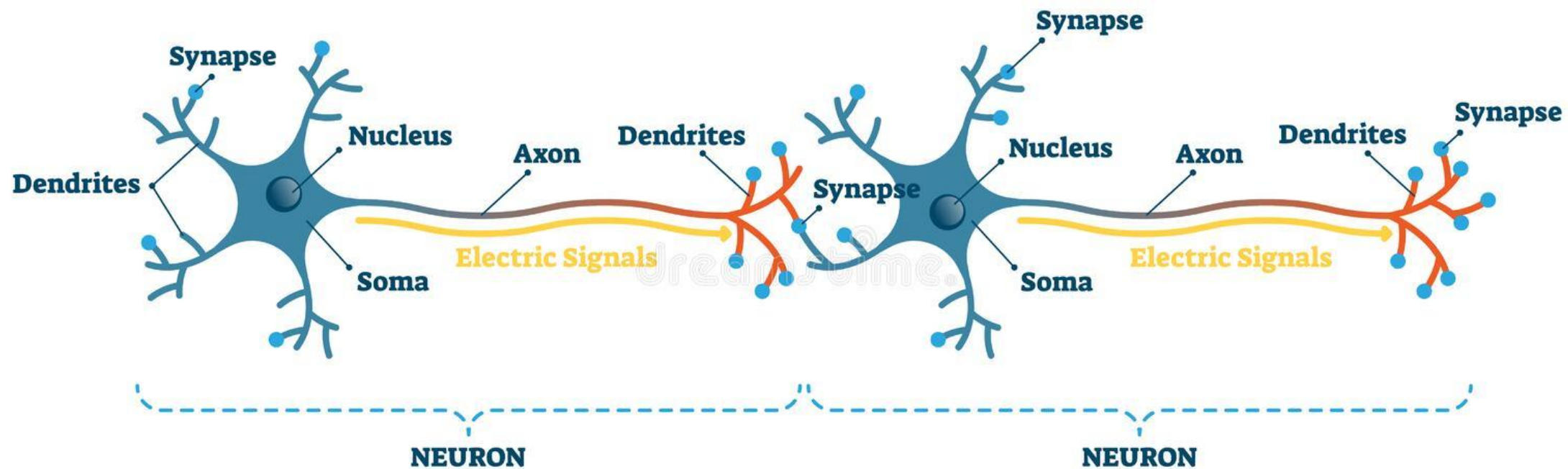
Motor Neuron



Sensory Neuron



Inter Neuron



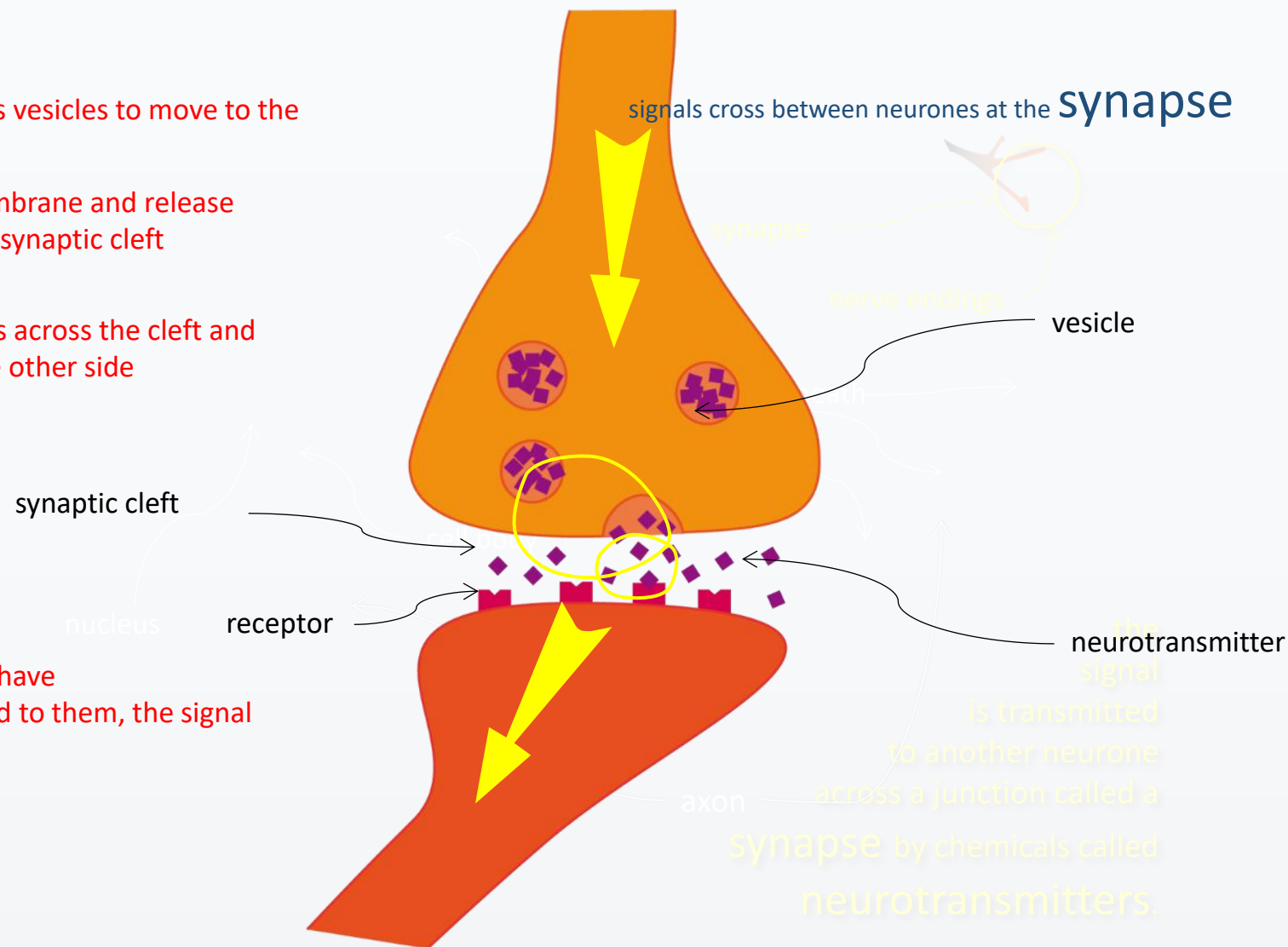


1 electrical impulse triggers vesicles to move to the synapse membrane

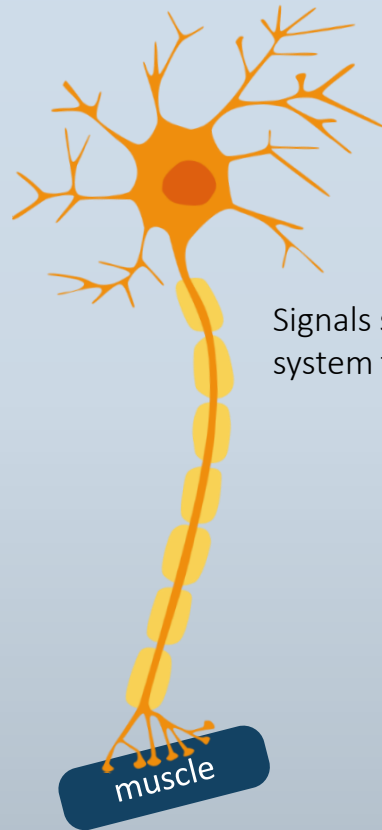
2 vesicles fuse with the membrane and release neurotransmitter into the synaptic cleft

3 neurotransmitter diffuses across the cleft and binds to receptors on the other side

4 Once enough receptors have neurotransmitters bound to them, the signal is transmitted...

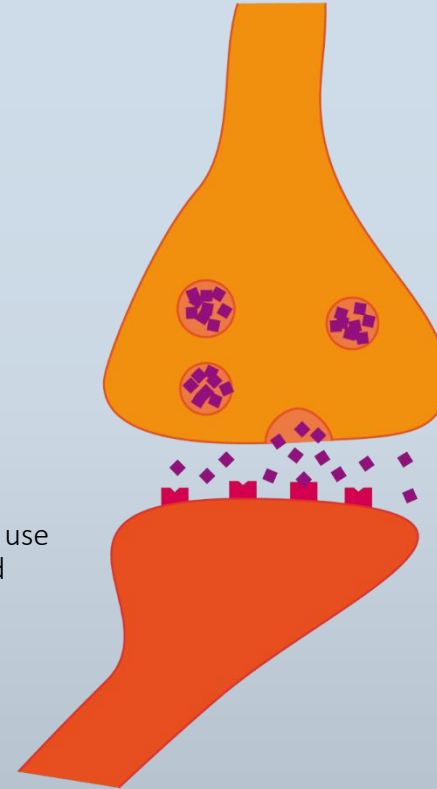


The point where your muscles and nervous system meet is called the
neuromuscular junction (NMJ)



Signals sent from your central nervous system to the NMJ tell muscles to move

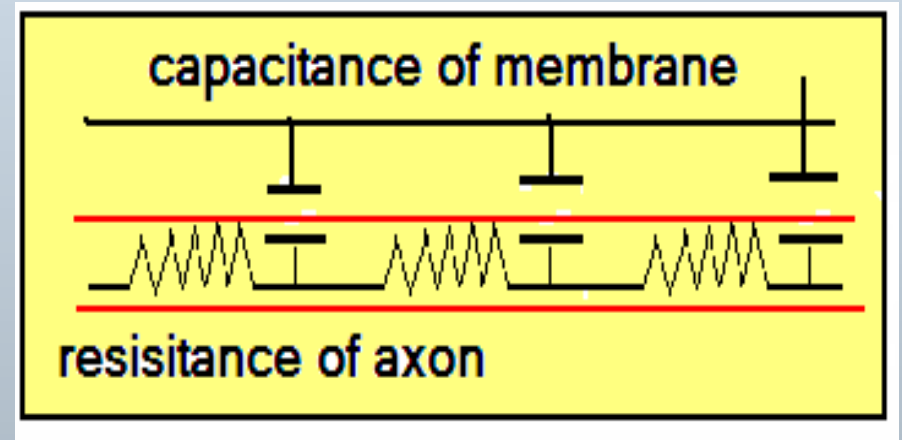
The synapses at the NMJ use a neurotransmitter called
acetylcholine

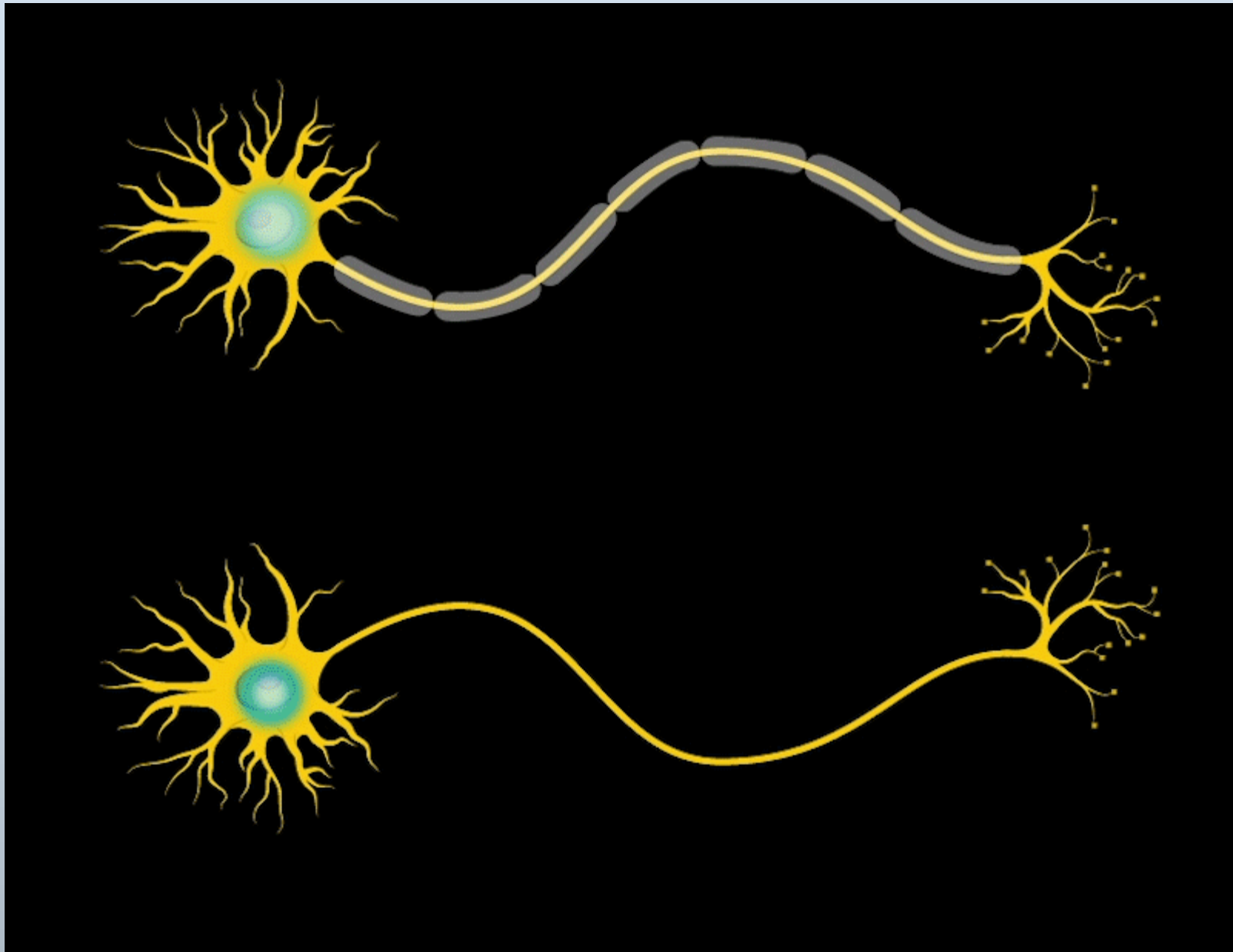


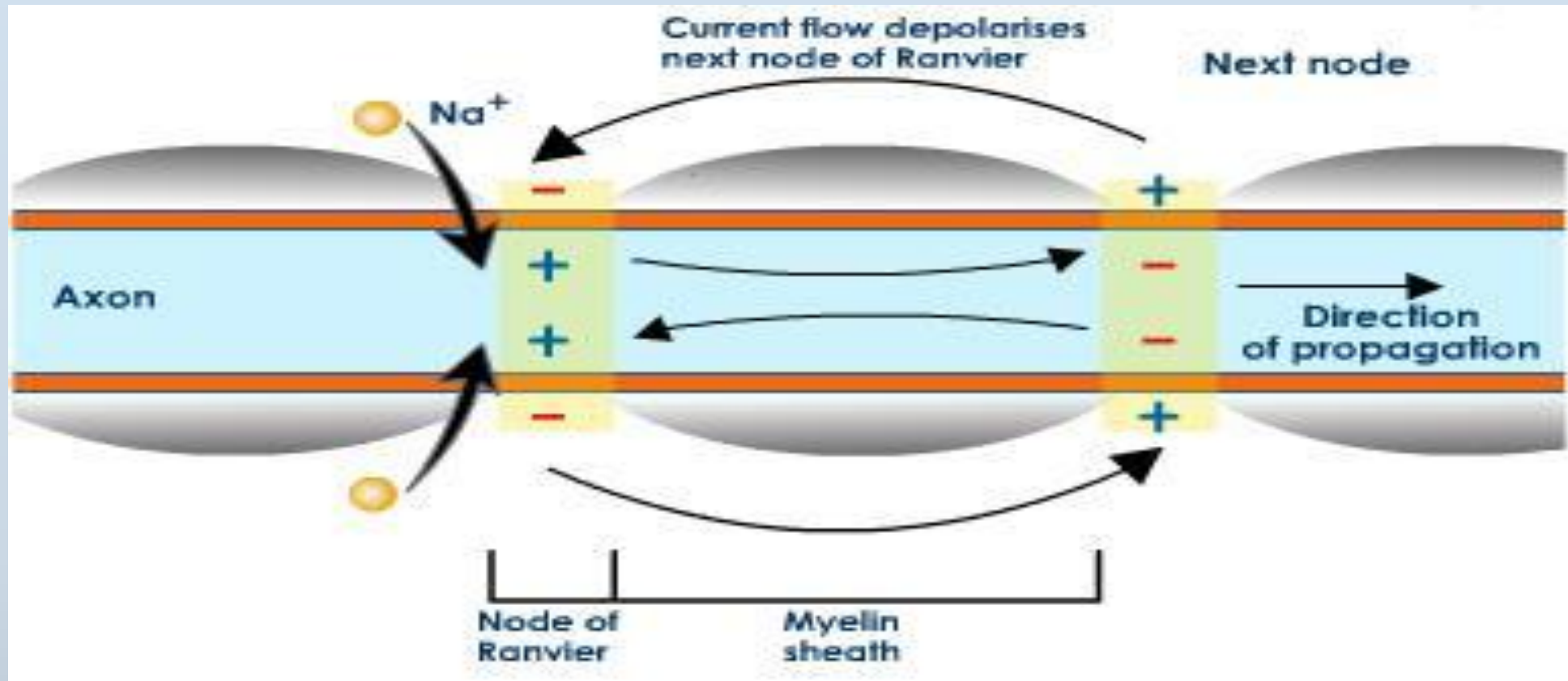
Nerve Fiber



- Two primary factor affect the speed of propagation of the action potential
 - Diameter
(larger diameter faster signal propagation)
 - Resistance $\propto$ cress sectional area
 - Capacitance $\propto$ cross sectional area
 - Myelination
 - Reduction of discharge in the cell membrane
 - Capacitance connected in series
 - Increasing of the electric insulator
 - Resistance connected in series
 - Increasing of the conduction velocity
 - Time $\propto 1/RC$
 - Unmyelinated axon = 0.5 -10 m/s
 - Myelinated axon up to 150 m/s







- No depolarisation occurs under the myelin
- Depolarisation only happens at the nodes ($0.5\mu\text{m}$)
- All the Na^+ channels are concentrated at the nodes.

- An impulse is triggered by local currents that depolarise the next bit of the membrane
- In myelinated nerves the triggering jumps from one node to the next
- Much quicker than depolarising all the membrane along the whole axon.



References

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